Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack engineer with 2+ years building **secure cloud-native applications**, **automation tooling**, and **reliable APIs** using React, Python, Java, SQL, Docker, Kubernetes, CI/CD, and AI-assisted workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Designed **multi-tenant backend services** on Azure SQL and PostgreSQL, optimizing schema and query paths to reduce dashboard latency from **1.2s to 350ms** for 500+ users in production workflows.
- Implemented **FastAPI orchestration pipelines** processing 5K+ monthly jobs with validation, retry logic, and reconciliation checks, improving operational reliability, incident recovery speed, and confidence in automated cloud releases.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Built **React + Flask full-stack features** handling 10K+ daily requests, translating ambiguous product requirements into secure APIs, reusable UI modules, and production-ready workflows through iterative Agile collaboration.
- Added **CI/CD quality gates**, automated regression checks, and concurrent processing safeguards, reducing turnaround time by **60%** while maintaining reliability for high-volume AI-assisted grading and feedback services.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Developed **Python evaluation pipelines** across 100K+ trajectory logs with reproducible experiments, structured metrics, and automated diagnostics, enabling faster debugging and stronger evidence-driven decisions for production system behavior.
- Partnered with engineers and researchers to convert analysis outputs into actionable implementation plans, improving technical communication, reducing iteration risk, and accelerating delivery of resilient AI-driven product capabilities.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL microservices** using Python/Java, AWS Glue, and S3, improving response times by **30%** while supporting secure, scalable integrations across distributed systems and enterprise data flows.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, boosting pipeline efficiency by **68%**, reducing technical debt by **20%**, and strengthening release quality in multi-environment deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **MERN application modules** with REST APIs and WebSockets supporting eight workflows and 2M+ synchronized records, improving scalability, real-time consistency, and maintainability for customer-facing operations and reporting.
- Automated **Docker and Jenkins deployment workflows**, cutting release time by **50%**, standardizing testing execution, and enabling safer, faster production rollouts under iterative Agile delivery schedules.

Skills

Languages: **Python**, **JavaScript/TypeScript**, **Java**, SQL, C++
Frontend: **React.js**, HTML5, CSS3, reusable component architecture
Backend: **Flask**, **FastAPI**, Node.js, REST APIs, microservices, server-side testing
Cloud/DevOps: **AWS**, Azure SQL, Docker, **Kubernetes**, Git, TeamCity-adjacent CI/CD practices, Jenkins
Reliability: observability-oriented debugging, logging/metrics mindset, incident triage, automated quality checks

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a **scalable full-stack platform** for 50K+ users and 1M+ transactions with secure integrations, real-time updates, and Kubernetes-backed deployment workflows that improved reliability for high-volume financial operations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Created a **schema-aware query assistant** supporting 10+ databases with validated SQL generation, structured execution outputs, and cross-platform tooling that accelerated developer productivity and reduced manual query debugging overhead.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*
- Implemented a **containerized cloud API** with automated validation, CI/CD release controls, and robust deployment checks, achieving **87%** processing accuracy while improving operational consistency in production environments.